Year 10 Mathematics

Number 1

1.1 Working with Fractions

Working with Fractions

STARTER · 5 MIN

Date: _____ **Title:** Working with Fractions

LEARNING OBJECTIVES

- Simplify fractions and multiply fractions
- Multiply and divide fractions and mixed numbers
- Add and subtract fractions and mixed numbers, and solve problems

STARTER *Answer in your book. No calculators.*

1. Find the HCF of 12 and 18.

2. Find the LCM of 4 and 6.

3. Write $2\frac{1}{3}$ as an improper fraction.

4. Write $\frac{17}{5}$ as a mixed number.

Starter — Answers

STARTER

Tick or correct in green pen

1. HCF of 12 and 18 = **6**

2. LCM of 4 and 6 = **12**

3. $2\frac{1}{3} = \frac{2 \times 3 + 1}{3} = \mathbf{\frac{7}{3}}$

4. $17 \div 5 = 3$ remainder 2, $\frac{17}{5} = \mathbf{3\frac{2}{5}}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can simplify fractions and multiply fractions	☐
SILVER	I can multiply and divide fractions and mixed numbers	☐
GOLD	I can add and subtract fractions and mixed numbers, and solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Fractions

INTRODUCTION · 3 MIN

Change mixed numbers to improper fractions first. Simplify every answer.

Simplify

Divide the top and the bottom by the HCF.

$$\frac{28}{42} = \frac{2}{3}$$

Multiply

Multiply the tops. Multiply the bottoms.

$$\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

Divide

Flip the second fraction, then multiply.

$$\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times \frac{2}{1} = 1\frac{1}{2}$$

Add and subtract

Make the denominators the same first.

$$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$

Examples 1 and 2 — Simplify and Multiply

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Simplify $\frac{18}{24}$

Example 2

Work out $\frac{2}{3} \times \frac{9}{10}$

Give your answer in its simplest form.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

The HCF of 18 and 24 is 6.

$$\frac{18 \div 6}{24 \div 6}$$

$$= \frac{3}{4}$$

Tip: check that 3 and 4 have no common factor.

Example 2

$$\frac{2 \times 9}{3 \times 10}$$

$$= \frac{18}{30}$$

$$= \frac{3}{5}$$

Tip: divide 18 and 30 by their HCF, 6.

Task 1

PRACTICE · 9 MIN

BRONZE

Give each answer in its simplest form

1 Simplify $\frac{12}{16}$

2 Simplify $\frac{30}{45}$

3 Work out $\frac{3}{4} \times \frac{2}{9}$

4 Work out $\frac{5}{6} \times \frac{3}{10}$

5 A class has 15 students. 6 of them are girls. What fraction of the class are girls?

★ Work out $\frac{4}{7}$ of 35

Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1 $\frac{12}{16} = \frac{12 \div 4}{16 \div 4} = \mathbf{\frac{3}{4}}$

2 $\frac{30}{45} = \frac{30 \div 15}{45 \div 15} = \mathbf{\frac{2}{3}}$

3 $\frac{3}{4} \times \frac{2}{9} = \frac{6}{36} = \mathbf{\frac{1}{6}}$

4 $\frac{5}{6} \times \frac{3}{10} = \frac{15}{60} = \mathbf{\frac{1}{4}}$

5 $\text{Girls} = \frac{6}{15} = \mathbf{\frac{2}{5}}$

★ $35 \div 7 = 5, \quad 5 \times 4 = \mathbf{20}$

Examples 3 and 4 — Mixed Numbers and Dividing

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Work out $1\frac{2}{3} \times 2\frac{1}{4}$

Example 4

A piece of wood is $4\frac{1}{2}$ m long. It is cut into pieces that are each $\frac{3}{4}$ m long.
How many pieces are there?

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$\frac{5}{3} \times \frac{9}{4}$$

$$= \frac{45}{12}$$

$$= \frac{15}{4}$$

$$= 3\frac{3}{4}$$

Tip: change mixed numbers to improper fractions first.

Example 4

$$\frac{9}{2} \div \frac{3}{4}$$

$$= \frac{9}{2} \times \frac{4}{3}$$

$$= \frac{36}{6}$$

$$= \mathbf{6 \text{ pieces}}$$

Tip: keep, change, flip.

Task 2

PRACTICE · 9 MIN

SILVER

Give each answer in its simplest form

1 Work out $\frac{3}{4} \div \frac{7}{8}$

2 Work out $6 \div 1\frac{1}{3}$

3 Work out $1\frac{3}{4} \times 2\frac{2}{3}$

4 Work out $3\frac{1}{3} \div 1\frac{1}{9}$

5 A jug holds $2\frac{1}{4}$ litres of juice. How many glasses of $\frac{3}{8}$ litre can be filled from the jug?

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $\frac{3}{4} \div \frac{7}{8} = \frac{3}{4} \times \frac{8}{7} = \frac{24}{28} = \mathbf{\frac{6}{7}}$

2 $6 \div 1\frac{1}{3} = 6 \div \frac{4}{3} = 6 \times \frac{3}{4} = \frac{18}{4} = \mathbf{4\frac{1}{2}}$

3 $1\frac{3}{4} \times 2\frac{2}{3} = \frac{7}{4} \times \frac{8}{3} = \frac{14}{3} = \mathbf{4\frac{2}{3}}$

4 $3\frac{1}{3} \div 1\frac{1}{9} = \frac{10}{3} \div \frac{10}{9} = \frac{10}{3} \times \frac{9}{10} = \mathbf{3}$

5 $2\frac{1}{4} \div \frac{3}{8} = \frac{9}{4} \times \frac{8}{3} = \frac{72}{12} = \mathbf{6\ glasses}$

Examples 5 and 6 — Add, Subtract and Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (3 marks)

Show that $3\frac{1}{3} - 1\frac{3}{4} = 1\frac{7}{12}$

Example 6 (3 marks)

Sara spends $\frac{1}{4}$ of her money on food and $\frac{1}{3}$ of her money on clothes.
What fraction of her money is left?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$\frac{10}{3} - \frac{7}{4}$$

$$= \frac{40}{12} - \frac{21}{12}$$

$$= \frac{19}{12}$$

$$= \mathbf{1\frac{7}{12}}$$

Tip: the common denominator of 3 and 4 is 12.

Example 6

$$\frac{1}{4} + \frac{1}{3} = \frac{3}{12} + \frac{4}{12} = \frac{7}{12}$$

$$1 - \frac{7}{12} = \frac{12}{12} - \frac{7}{12}$$

$$= \mathbf{\frac{5}{12}}$$

Tip: the whole amount is 1.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1 Work out $\frac{3}{8} + \frac{7}{12}$

(2 marks)

2 Work out $4\frac{3}{5} - 2\frac{1}{2}$

(2 marks)

3 Show that $1\frac{3}{4} + 2\frac{5}{6} = 4\frac{7}{12}$

(3 marks)

4 Show that $2\frac{1}{3} \times 1\frac{4}{5} = 4\frac{1}{5}$

(3 marks)

5 Tom has $2\frac{1}{2}$ kg of rice. He uses $\frac{3}{4}$ kg on Monday and $1\frac{1}{3}$ kg on Tuesday.

How much rice is left?

(3 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $\frac{3}{8} + \frac{7}{12} = \frac{9}{24} + \frac{14}{24} = \mathbf{\frac{23}{24}}$

2. $4\frac{3}{5} - 2\frac{1}{2} = \frac{46}{10} - \frac{25}{10} = \frac{21}{10} = \mathbf{2\frac{1}{10}}$

3. $1\frac{3}{4} + 2\frac{5}{6} = \frac{21}{12} + \frac{34}{12} = \frac{55}{12} = \mathbf{4\frac{7}{12}}$

4. $2\frac{1}{3} \times 1\frac{4}{5} = \frac{7}{3} \times \frac{9}{5} = \frac{63}{15} = \frac{21}{5} = \mathbf{4\frac{1}{5}}$

5. Rice left $= 2\frac{1}{2} - \frac{3}{4} - 1\frac{1}{3} = \frac{30}{12} - \frac{9}{12} - \frac{16}{12} = \mathbf{\frac{5}{12}}$ kg

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $\frac{24}{36}$

SILVER Work out $1\frac{1}{2} \div \frac{3}{4}$

GOLD Work out $2\frac{1}{2} \times \frac{2}{5} + \frac{3}{4}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE

$$\frac{24}{36} = \frac{24 \div 12}{36 \div 12} = \mathbf{\frac{2}{3}}$$

SILVER

$$1\frac{1}{2} \div \frac{3}{4} = \frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = \mathbf{2}$$

GOLD

$$2\frac{1}{2} \times \frac{2}{5} + \frac{3}{4} = \frac{5}{2} \times \frac{2}{5} + \frac{3}{4} = 1 + \frac{3}{4} = \mathbf{1\frac{3}{4}}$$

Exam Practice — Fractions of Shapes and Amounts

EXAM PRACTICE · 10 MIN

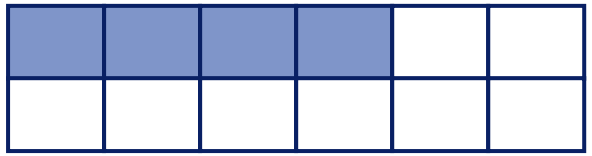
BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(3 marks)

The diagram shows a shape made of 12 equal rectangles.



(a) What fraction is shaded? (1)

.....

(b) How many more must be shaded so that $\frac{5}{6}$ of the shape is shaded? (2)

.....

Question 2

(4 marks)

(a) Work out $\frac{3}{4}$ of 56 kg. (2)

..... kg

(b) Show that (2)

$$\frac{5}{8} \times \frac{4}{15} = \frac{1}{6}$$

.....

Exam Practice — Mixed Numbers

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(3 marks)

Show that

$$2\frac{2}{3} \times 1\frac{1}{8} = 3$$

Question 4

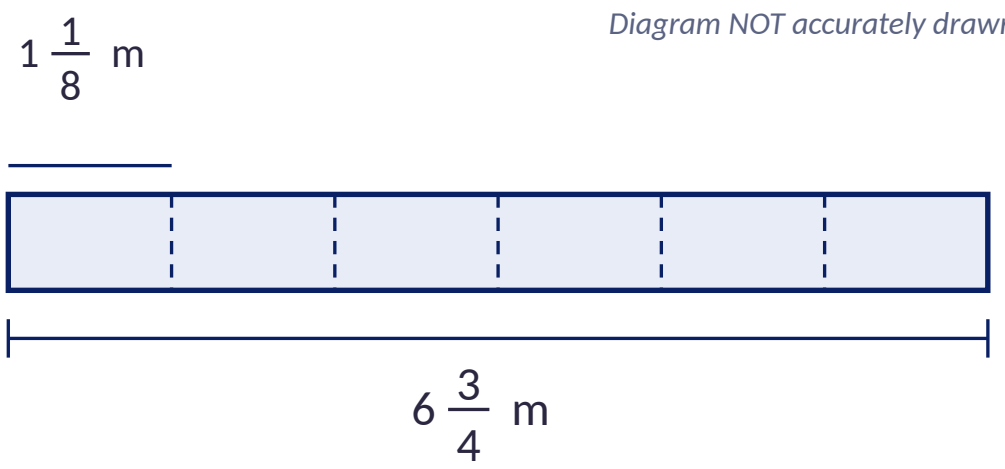
(3 marks)

A plank is cut into equal pieces.

The diagram shows the lengths.

How many pieces are cut?

Diagram NOT accurately drawn



..... pieces

Exam Practice — Fraction Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

Show that

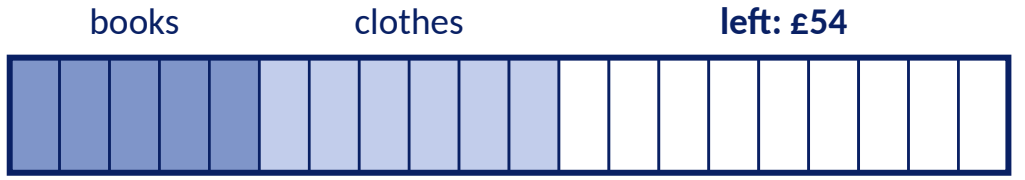
$$3\frac{1}{3} - 1\frac{4}{5} = 1\frac{8}{15}$$

Question 6

(4 marks)

Amira spends $\frac{1}{4}$ of her money on books and $\frac{2}{5}$ of the rest on clothes.

She has £54 left. How much did she have at the start?



..... £

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $\frac{4}{12} = \frac{1}{3}$ — **B1**

(b) $\frac{5}{6}$ of 12 = 10 — **M1**

10 − 4 = 6 more — **A1**

Tip: find the fraction of the whole shape first.

Question 2

(a) $56 \div 4 \times 3$ — **M1**

= 42 kg — **A1**

(b) $5 \times 4 = 20$, $8 \times 15 = 120$ — **M1**

$\frac{20}{120} = \frac{1}{6}$ **(shown)** — **A1**

Tip: in 'show that', write every step.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$\frac{8}{3} \times \frac{9}{8}$ M1

$= \frac{72}{24}$ M1

= 3 (shown) A1

Tip: change to improper fractions first.

Question 4

$\frac{27}{4} \div \frac{9}{8}$ M1

$= \frac{27}{4} \times \frac{8}{9}$ M1

= 6 pieces A1

Tip: divide by a fraction = × its reciprocal.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$\frac{10}{3} - \frac{9}{5}$ — M1

$= \frac{50}{15} - \frac{27}{15}$ — M1

$= \mathbf{\frac{23}{15} = 1\frac{8}{15}}$ — A1

Tip: use a common denominator of 15.

Question 6

clothes $= \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$ — M1

left $= 1 - \frac{1}{4} - \frac{3}{10} = \frac{9}{20}$ — M1

$\frac{9}{20}$ of total = £54 — M1

total = £120 — A1

Tip: clothes is a fraction of the REST, not the total.